

This lecture began by recalling half of the material from lecture 14 since we had to make the latter online but it had a new and fundamental object of study to us, namely vector spaces; in these notes we omit what was recalled and only begin with what is not in lecture 14 notes.

Definition 29. Let V be a vector space over $F = \mathbb{R}$ or $\mathbb{C}$ and S a subset of V . We say S is:

- closed under addition if for all $\mathbf{u}, \mathbf{v} \in V$, if $\mathbf{u}, \mathbf{v} \in S$ then $\mathbf{u} + \mathbf{v} \in S$.
- closed under scalar multiplication if for all $\mathbf{u} \in V$, and $\lambda \in F$ if $\mathbf{u} \in S$ then $\lambda \mathbf{u} \in S$.

Examples Consider $\mathbb{R}^2$ (with component-wise addition and scalar multiplication) and let $S = \{(x, y) \in \mathbb{R}^2 | x \geq 0 \text{ and } y \geq 0\}$ (the first quadrant in the Cartesian plane). We have that S is closed under addition for if we take $(x_1, y_1), (x_2, y_2) \in S$ we must have $x_1 \geq 0$ and $x_2 \geq 0$ and thus $x_1 + x_2 \geq 0$, similarly $y_1 + y_2 \geq 0$ and so $(x_1, y_1) + (x_2, y_2) \in S$. However it is not closed under scalar multiplication since $(2023, 2095) \in S$ but $-(2023, 2095) = (-2023, -2095) \notin S$.

Consider again $\mathbb{R}^2$ (with component-wise addition and scalar multiplication) and let $T = \{(x, y) \in \mathbb{R}^2 | x \geq 0 \text{ and } y \geq 0\} \cup \{(x, y) \in \mathbb{R}^2 | x \leq 0 \text{ and } y \leq 0\}$ (the first together with the third quadrant in the Cartesian plane). Check that T is closed under scalar multiplication (you may need to consider different cases), however T is not closed under addition - take $(-2, -1), (1, 2) \in T$ for example.

Subspaces

Definition 30. A subset S of a vector space V (over F) is called a subspace if

- S is non-empty;
- S is closed under addition;
- S is closed under scalar multiplication.

Examples (1) The trivial vector space $V = \{0\}$ has only one subspace, namely itself.

(2) Let $V = M_{m \times n}(\mathbb{R})$ (with component-wise addition and scalar multiplication). Your sets $\mathcal{L}$, $\mathcal{R}$ and $\mathcal{D}$ of lower, upper and diagonal matrices, respectively are subspaces (check this).